metaDMG-cpp file formats

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This document describes the file formats currently produced and consumed by metaDMG-cpp. Some legacy formats are retained for backwards compatibility and are marked as such.

- .bdamage.gz
- .lca (or .lca.gz)
- .stat (or .stat.gz)
- .dfit (or .dfit.gz)
- .boot.stat (or .boot.stat.gz)
- .aggregate.stat (or .aggregate.stat.gz)
- .rlens.gz
- .usedreads.bam and .famoutreads.bam

1 bdamage format

bdamage files are files that contain counts of mismatchs conditional on strand and cycle (position within read). These are generated with metadamage lca or metadamage getdamage. The first 8 bytes magic number determines which bdamage version. If no magic number is present then version0 is assumed.

1.1 version 0

First version of the bdamage file is a single bgzf compressed file. MAXLENGTH occurs once in the beginning of the file, followed by succesive blocks of data[1-8]. Block[3-5] indicates the actual mismatch counts for the forward which we have MAXLENGTH times. Block[6-8] indicates the actual mismatch counts for the reverse strand which will also occur MAXLENGTH times.

1.2 version 1

Second iteration of the bdamage file as magicnumber[8] = "bdam1".

Col	Field	Type	Brief description
0	MAXLENGTH	int	Number of cycles
1	ID	int	Id for mismatch type[1]
2	NREADS	int	Number of reads used supporting the mismatch matrix
3	1	float[16]	mismatch rate for first cycle from the 5prime
4	i	float[16]	mismatch rate for the i'th cycle from the 5prime
5	MAXLENGTH	float[16]	mismatch rate for the last cycle from the 5prime
6	1	float[16]	mismatch rate for first cycle from the 3prime
7	i	float[16]	mismatch rate for the i'th cycle from the 3prime
8	MAXLENGTH	float[16]	mismatch rate for the last cycle from the 3prime

Table 1: Content of bdamage.gz file. Note[1] This is either the taxidID or the referenceID relative to the SAM/BAM header for single species resequencing projects or it is the *taxid* if output has been generated with metadamage lca 3) Order is given by AA,AC,AG,AT,CA,CC,CG,CT,GA,GC,GG,GT,TA,TC,TG,TT, with first base indicatting reference nucleotide and second base indicating observed nucleotide

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2 .lca

This section describes the test output generated by a metadamage lca subfunctionality and contains information at the readlevel regarding both taxonomic information and statistics pertaining usefull readinformation.

The first line is a comment beginning with a hashtag and normally contains the command used to generate the file. The exact compression suffix depends on the calling workflow.

The first non-comment line is a tab-separated header: `queryid seq len naln nspec dustscore gc lca taxa_path`.
Each following line is one read with fixed columns. The `lca` and `taxa_path` fields are structured as `taxid:"name":"rank"` blocks. The complete specification is shown below.

Col	Brief description	
1	queryid	read identifier
2	seq	read sequence
3	len	read length
4	naln	number of alignments used for LCA
5	nspec	number of unique species represented among the alignments
6	dustscore	DUST low-complexity score for the read sequence
7	gc	GC fraction of the read sequence
8	lca	`taxid:"name":"rank"` for the inferred LCA node
9	taxa_path	semicolon-separated `taxid:"name":"rank"` from LCA to root

Table 2: Content of a .lca file. Fields are tab-separated. Quotes in `lca` and `taxa_path` preserve names/ranks that may contain spaces. The number of tab-separated columns is fixed across reads.

3 .stat

Very simple tabseperated flatfile

1. taxid
2. Number of supporting reads
3. Mean lengths of supporting reads
4. Variance of the lengths of the supporting reads
5. Mean gccontent of supporting reads
6. Variance of the gccontent of supporting reads
7. Mean dustscore of supporting reads
8. Variance of the dustscore of supporting reads
9. Mean nspec value of supporting reads
10. Variance of the nspec value of supporting reads
11. name of lca in quotes (if relevant, otherwise NA)
12. name of taxomic level of lca in quotes (if relevant, otherwise NA)

The current header is:

```
taxid nreads mean_len var_len mean_gc var_gc mean_dust var_dust
mean_nspec var_nspec lca rank
```

4 Read-selection BAM files

The LCA command writes BAM files with the same header as the input alignment file when the corresponding options are enabled. With output prefix `X`:

- `X.usedreads.bam` contains alignments used for damage analysis.
- `X.famoutreads.bam` contains alignments from read groups whose inferred LCA is at family rank or a more specific rank. This output is controlled by `--famout` and is enabled by default.

5 .dfit

The output format depends on the supplied parameters and run mode. It can contain the per file, the per reference or the per species estimate of damage.
The current output starts with the following columns, independent of run mode:

```
taxid A q c phi llh ncall sigmaD_old Zfit_old sigmaD_new Zfit_new
A_b q_b c_b phi_b A_CI_l A_CI_h q_CI_l q_CI_h c_CI_l c_CI_h
phi_CI_l phi_CI_h
```

The bootstrap and confidence-interval columns are present when bootstrap summaries are enabled. They are followed by repeated per-cycle forward and reverse-strand fields. For each cycle i, the current fields are `fwK`i, `fwN`i, `fwf`i, `fwdx`i, `fwdxConf`i and the corresponding `bwK`i, `bwN`i, `bwf`i, `bwdx`i, `bwdxConf`i fields. The exact number of cycles is determined by the input mismatch matrix.
Non-boostrapping optimization:

Column name	Brief description
	–showfits 0
id	identifier see paragraph for details
A	Dfit statistic. Damage at position one, taking into account offset
q	per cycle decrease
c	background substitution rate or noise baseline
ϕ	Variance between beta-binomial and binomial model
llh	Likelihood for our MLE
ncall	Number of optimization calls used for obtaining our MLE
$\sigma_{D,old}$	Legacy Z value
Zfit_old	Legacy significance
$\sigma_{D,new}$	Current effective-N Z value
Zfit_new	Current effective-N significance
	–showfits 1, i signifies position inferred from *.bdamage.gz*
$fwdxi$	damage estimates forward strand
$fwdConfi$	forward strand confidence interval ±
$bwdxi$	damage estimates backward strand
$bwdConfi$	backward strand confidence interval ±
	–showfits 2, i signifies position inferred from *.bdamage.gz*
$fwKi$	Number of deamination substitution (C→T) observations forward strand
$fwNi$	Total number of C for forward strand
$fwdxi$	Damage estimates forward strand
$fwfi$	Calculated damage frequency $fwKi/fwNi$
$fwdConfi$	forward strand confidence interval ±
$bwKi$	Number of deamination substitution (G→A for ds, C→T for ss) observations forward strand
$bwNi$	Total number of G for forward strand (ds) and C (ss)
$bwdxi$	Damage estimates backward strand
$bwfi$	Calculated damage frequency $fwKi/fwNi$
$bwdConfi$	backward strand confidence interval ±

Table 3: Current content of a .dfit file. The per-cycle fields are repeated for the 5' and 3' ends as applicable.

Boostrapping optimization:
Providing *-nbootstrap* > 1 numerical optimizations of the binomial distribution will also be conducted using bootstrapping methods

Depending on which parameters and runmode (local,global or lca)that was supplied to both *getdamage* and *dfit*, the content of the id will be different. If a bamfile is supplied (with –bam) then each line will be the information associated with the different refids in the bam file and the id will be the referenceids from the bam file. The case scenario for this would be either obtaining perchromosome estimates of damage or per reference damage which could be relevant for metagenomic studies. If user are computing the damagesignal in the context of the lca. Then the id column will contain the taxid. If -names has been supplied to the dfit program, then the id column will the taxid:*scientific name*. If -nodes has not been defined the dfit file will only contain information for the observed references. If -nodes has been defined the program will aggregate the summary statitics for the internal nodes.

6 .boot.stat

When the *dfit* command is run with *nbootstrap* > 1 and *–printboot 1*, bootstrap values are stored in a separate `.boot.stat` file. The current header is:

```
taxid A_b q_b c_b phi_b sigmaD_old_b Zfit_old_b sigmaD_new_b Zfit_new_b
```

Each row represents one bootstrap draw for one taxon or reference.

7 .aggregate.stat

This format aggregates the information stored in the LCA *.stat* format. The current header is:

```
taxid name rank nalign nreads mean_rlen var_rlen mean_gc var_gc
mean_dust var_dust mean_nspec var_nspec lca taxa_path
```

If dfit output is supplied to aggregate, its columns are appended after `taxa_path`.

8 .rlens.gz

Readlength distribution. Distribution is count of alignments of specific readlengths. Depending on runmode there might be multiple groups as taxid/refs and there will be group specific distributions.

8.1 Legacy layout

Distributions for different groups are split by newlines. First entry on each line is the identifer (chromosomename,taxid). The remaining entries are the number of times we have observed an alignment of length (columnnumber). Notice that the first 30 column of counts is likely to be zero since reads shorter than 30basepairs are normally discarded.

8.2 Current layout: compact format

Each distribution occupies one line. The header is `id trlen:count`; each following line contains an ID followed by tab-separated `length:count` pairs. This is the default when `--rlens_flat_out 0` is used. Example can be seen below. This is the default option that should be enforcable by `--rlens_flat_out 0`. Output below is clipped at rightside and at bottom. Example shown is from one of the testfiles, and has been generated by `cut -c1-80 test/output/test_getdamage_local.rlens|head`

```
id rlen:count
0 35:33 36:29 37:24 38:38 39:24 40:29 41:36 42:28 43:27 44:36 45:29 46:23 47:32...
1 35:103 36:106 37:102 38:111 39:109 40:126 41:125 42:164 43:154 44:133 45:152...
2 35:35 36:44 37:28 38:41 39:33 40:36 41:41 42:33 43:34 44:43 45:41 46:36 47:40...
3 35:36 36:39 37:25 38:37 39:29 40:34 41:39 42:33 43:34 44:43 45:41 46:33 47:39...
4 35:100 36:98 37:95 38:112 39:107 40:136 41:127 42:161 43:157 44:138 45:134 46:...
5 35:92 36:118 37:110 38:125 39:129 40:168 41:158 42:164 43:195 44:180 45:177 46...
6 35:95 36:130 37:111 38:125 39:129 40:178 41:167 42:148 43:196 44:193 45:185 46...
7 35:138 36:145 37:131 38:125 39:147 40:179 41:190 42:205 43:196 44:201 45:220 4...
8 35:95 36:132 37:113 38:137 39:133 40:182 41:162 42:176 43:196 44:190 45:189 46...
...
```

8.3 Current layout: flat format

For easier machine parsing, use `--rlens_flat_out 1` or `-z 1` where supported. The header is `ID trlen`
`tcount`, followed by one row per ID and read length.

```
IFILE=./data/f570b1db7c.dedup.filtered.rname.bam
../metaDMG-cpp getdamage -r 1 -l 35 -p 5 -o output/test_getdamage_local ${IFILE} -z 1
gunzip -c output/test_getdamage_local.rlens.gz|head -n20
##below is the output from the last command above
ID rlen count
0 35 33
0 36 29
0 37 24
0 38 38
0 39 24
0 40 29
0 41 36
0 42 28
0 43 27
0 44 36
0 45 29
0 46 23
0 47 32
0 48 23
0 49 25
0 50 17
0 51 16
0 52 19
0 53 17
```

Column name	Brief description
	–showfits 0
id	identifier see paragraph for details
A	Dfit statistic. Damage at position one, taking into account offset
q	per cycle decrease
c	background substitution rate or noise baseline
ϕ	Variance between beta-binomial and binomial model
llh	Likelihood for our MLE
ncall	Number of optimization calls used for obtaining our MLE
$\sigma_{D,old}$	Legacy Z value
Zfit_old	Legacy significance
$\sigma_{D,new}$	Current effective-N Z value
Zfit_new	Current effective-N significance
A_b	Dfit statistic from bootstrap estimate. Damage at position one, taking into account offset
q_b	per cycle decrease from bootstrap estimate
c_b	background substitution rate or noise baseline from bootstrap estimate
ϕ_b	Variance between beta-binomial and binomial model from bootstrap estimate
A_CI_l	Lower bound of CI for A estimate calculated from all bootstrap values
A_CI_h	Upper bound of CI for A estimate calculated from all bootstrap values
q_CI_l	Lower bound of CI for q estimate calculated from all bootstrap values
q_CI_h	Upper bound of CI for q estimate calculated from all bootstrap values
c_CI_l	Lower bound of CI for c estimate calculated from all bootstrap values
c_CI_h	Upper bound of CI for c estimate calculated from all bootstrap values
ϕ_CI_l	Lower bound of CI for ϕ estimate calculated from all bootstrap values
ϕ_CI_h	Upper bound of CI for ϕ estimate calculated from all bootstrap values
	–showfits 1, *i* signifies position inferred from *.bdamage.gz*
$fwdxi$	damage estimates forward strand
$fwdConfi$	forward strand confidence interval ±
$bwdxi$	damage estimates backward strand
$bwdConfi$	backward strand confidence interval ±
	–showfits 2, *i* signifies position inferred from *.bdamage.gz*
$fwKi$	Number of deamination substitution (C→T) observations forward strand
$fwNi$	Total number of C for forward strand
$fwdxi$	Damage estimates forward strand
$fwfi$	Calculated damage frequency $fwKi/fwNi$
$fwdConfi$	forward strand confidence interval ±
$bwKi$	Number of deamination substitution (G→A for ds, C→T for ss) observations forward strand
$bwNi$	Total number of G for forward strand (ds) and C (ss)
$bwdxi$	Damage estimates backward strand
$bwfi$	Calculated damage frequency $fwKi/fwNi$
$bwdConfi$	backward strand confidence interval ±

Table 4: Content of a .dfit file. The per-cycle fields are repeated for each cycle, first from the 5énd and then from the 3énd.

Col	Brief description	
1	taxid	The lca taxomic id (integer)
2	name	The lca taxonomic name(string)
3	rank	The lca taxonomic rank/level (string)
4	nalign	The aggregated number of alignments used for inferring the lca across the taxonomic levels
5	nreads	The aggregated number of reads for inferring the lca across the taxonomic levels
6	mean_rlen	The weighted mean of read length when transvering through the taxonomic levels
7	var_rlen	The pooled variance of read length when transvering through the taxonomic levels
8	mean_gc	The weighted mean of gc content when transvering through the taxonomic levels
9	var_gc	The pooled variance of gc content when transvering through the taxonomic levels
10	mean_dust	The weighted mean of dust score when transvering through the taxonomic levels
11	var_dust	The pooled variance of dust score when transvering through the taxonomic levels
12	mean_nspec	The weighted mean of nspec when transvering through the taxonomic levels
13	var_nspec	The pooled variance of nspec when transvering through the taxonomic levels
14	lca	The lca rank
15	taxa_path	The taxonimcal path from lca to root

Table 5: Content of a .aggregate.stat file. Column 14 contains the LCA descriptor and column 15 contains the semicolon-separated path to the root.